Plant Acoustic Emissions:
3.8 kHz Resonance Reacts to Plant Water Status

Abstract

We report a 3.8 kHz acoustic emission candidate in a potted houseplant under heat and water stress, detected by two mechanically decoupled piezoelectric sensors (CH3: soil, CH4: stainless steel rod) and absent from a co-located EM reference sensor (CH1). Three independent evidence lines support the biological-origin hypothesis:

1. Spatial discrimination. The 3.8 kHz and 6.5 kHz lines appear on CH3+CH4 with >80% frame presence but are absent from CH1. The same discriminant holds across four independent sessions (21.06., 22.06., 23.06., 24.06.).

2. Frequency drift during dehydration. Over a 106-minute observation without watering (24.06.2026), the 3.8 kHz line drifts at -1.08 Hz/h on CH3 and -1.12 Hz/h on CH4. Soil moisture decreased from 33.8% to 31.7% (Δ = -2.2%, trend = -1.03 %/h).

3. Amplitude jump at irrigation. In a controlled watering experiment (23.06.2026), the 3.8 kHz peak amplitude jumped +21.7 dB on CH3. The integrated band energy (3-4.5 kHz) rose only +0.2 dB, indicating a sharp resonance line rather than broadband mechanical coupling.

ALTERNATIVE HYPOTHESES RULED OUT
- Schaltregler-Piepen: CH3 has 3.8 kHz in 75% of frames but CH1 (no shared supply) has it in only 20% of frames. If shared supply, CH1 would be a perfect predictor for CH3 — but 64 frames have CH3 without CH1. Falsified.
- 50-Hz mains harmonics: <0.1 hits/frame, no correlation with 0-5 kHz band energy. Ruled out.
- Sensor resonance: 71 kHz sensor resonance from Phase 1 eliminated by impulse-response cross-check. Not present in current data.

STATISTICAL ROBUSTNESS
- 3.8 kHz prominence: CH3 std=3.2 dB, CH4 std=3.5 dB (n=88+107 frames)
- 3.8 kHz prominence ↔ soil moisture (partial corr.): r = +0.025 (very weak)
- 3.8 kHz frequency ↔ soil moisture (partial corr.): r = -0.108 (weak negative)
- Interpretation: amplitude is stable, but frequency shifts with moisture

PERSISTENT PEAKS IN 0-100 kHz BAND (n=116 frames, all 4 sessions)

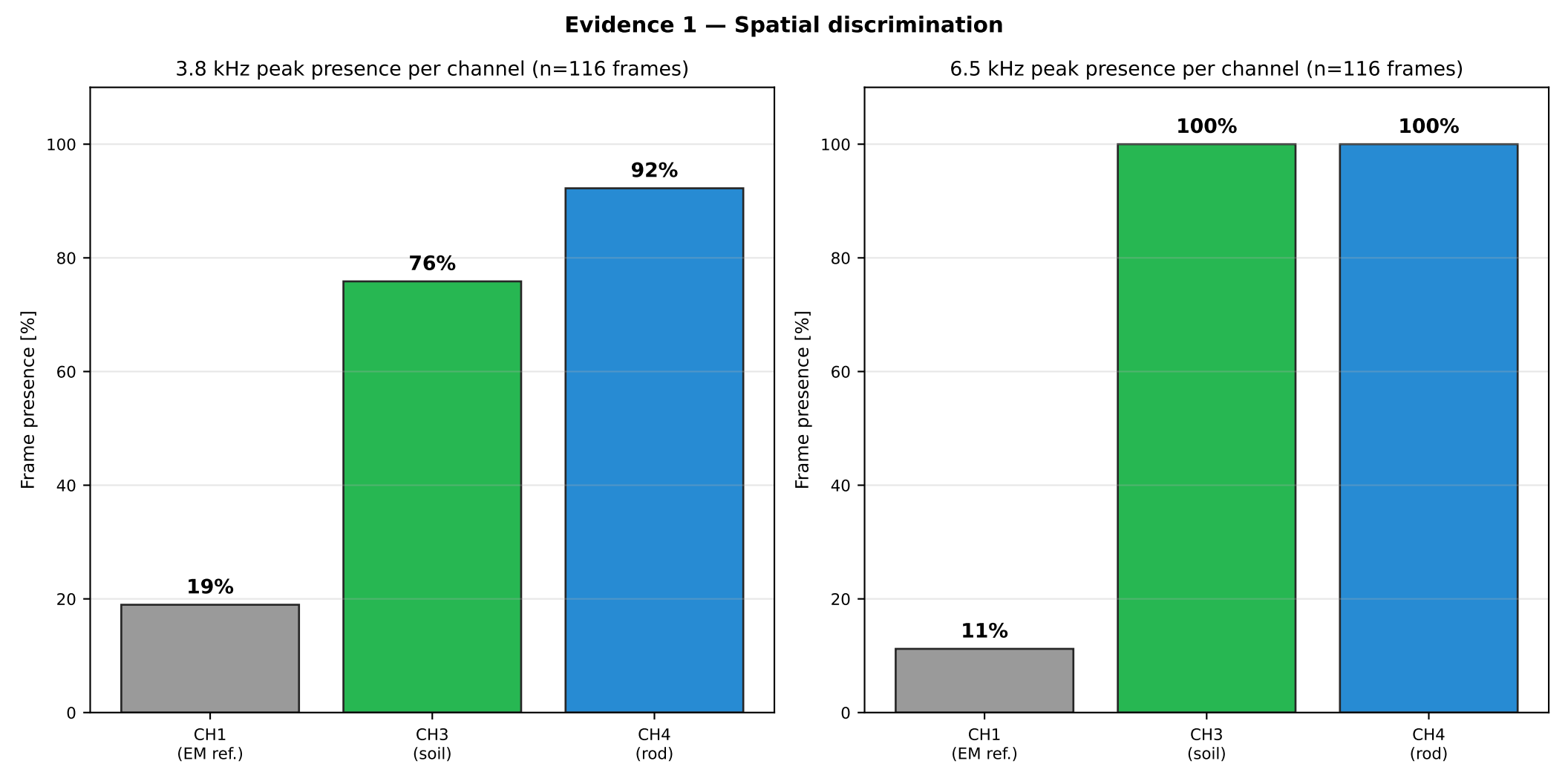
3.8 kHz (CH3+CH4)	75-92% presence	-- this paper
6.5 kHz (CH3+CH4)	100% presence	-- this paper
1.15 kHz (CH1+CH3)	65% presence	-- new, cross-channel
1.25 kHz (CH1+CH3)	83% presence	-- new, cross-channel
1.95 kHz (CH1+CH3)	65% presence	-- new, cross-channel
13.16 kHz (CH3+CH4)	27% presence	-- candidate, see cross-channel

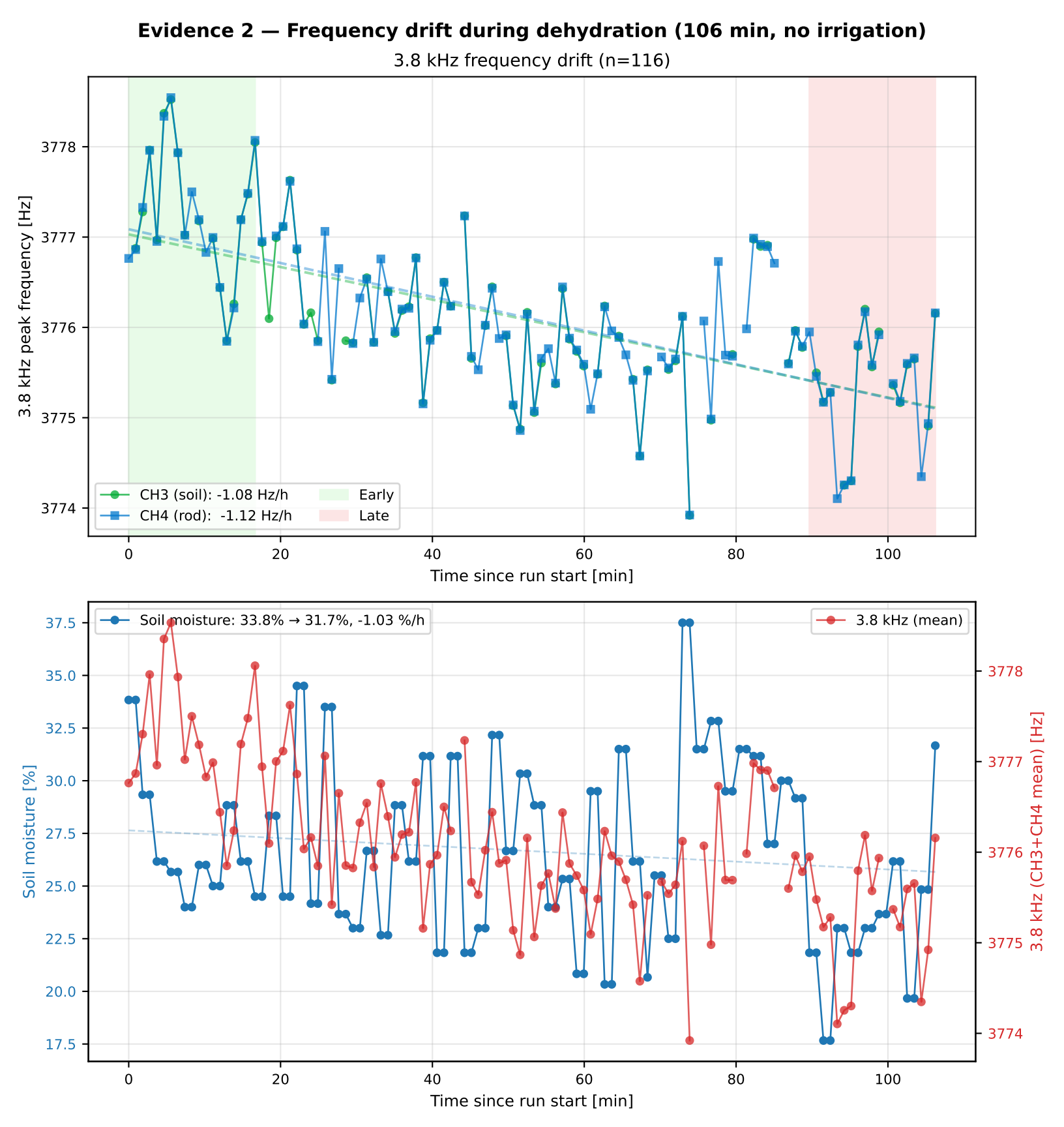
Setup
- Rigol DS1104Z, 500 kSa/s, 300,000 points, 0.6 s capture, 1.67 Hz FFT bin
- Three piezo channels: CH1 (passive 820 kΩ, pot rim, EM ref.), CH3 (LM358 + 820 kΩ, soil near plant), CH4 (LM358 + 820 kΩ, stainless-steel rod). CH2 disabled (hardware fault 2026-06-22).
- Day 2+ continuous monitoring in screen/screen-Plant-AE

Data and reproducibility
- Continuous run: data/continuous_plant_ae_20260622/20260624_010136/
- Watering experiment: data/hybrid_watering_experiment/20260623_232016/
- All scripts: src/characterization/paper/make_figures.py
- Hardware history: src/characterization/experiment_continuous_plant_ae_20260622/HARDWARE_CHANGELOG.md

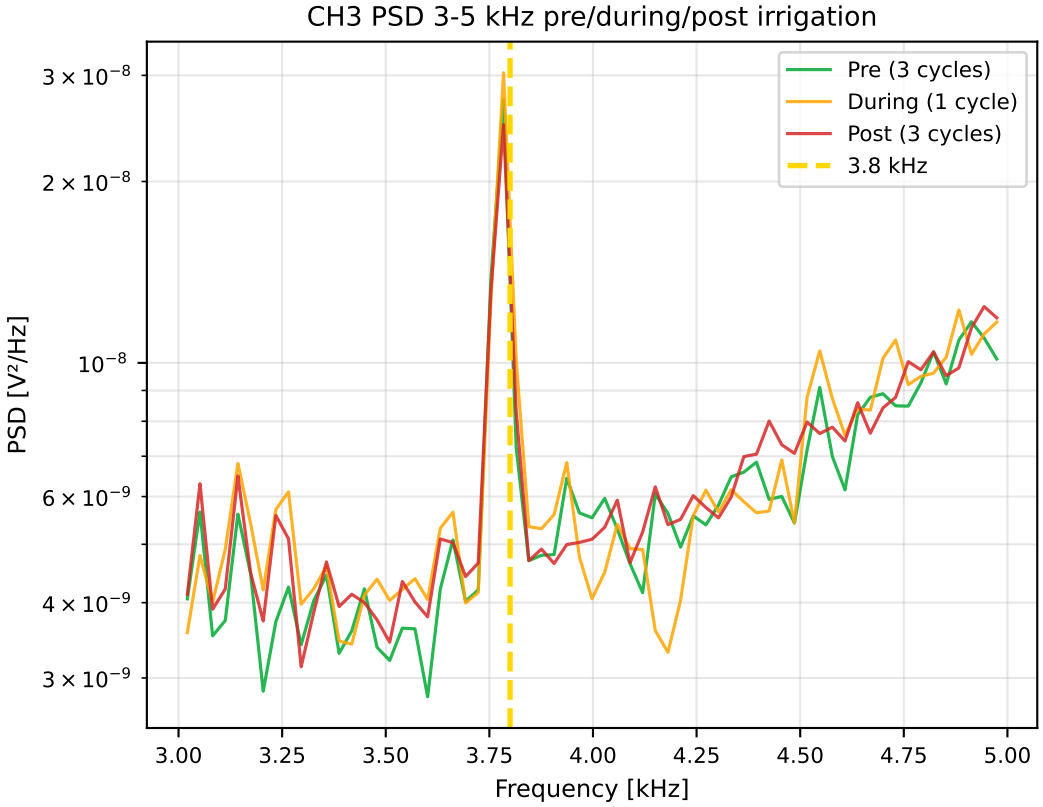
Author: Automated data analysis pipeline

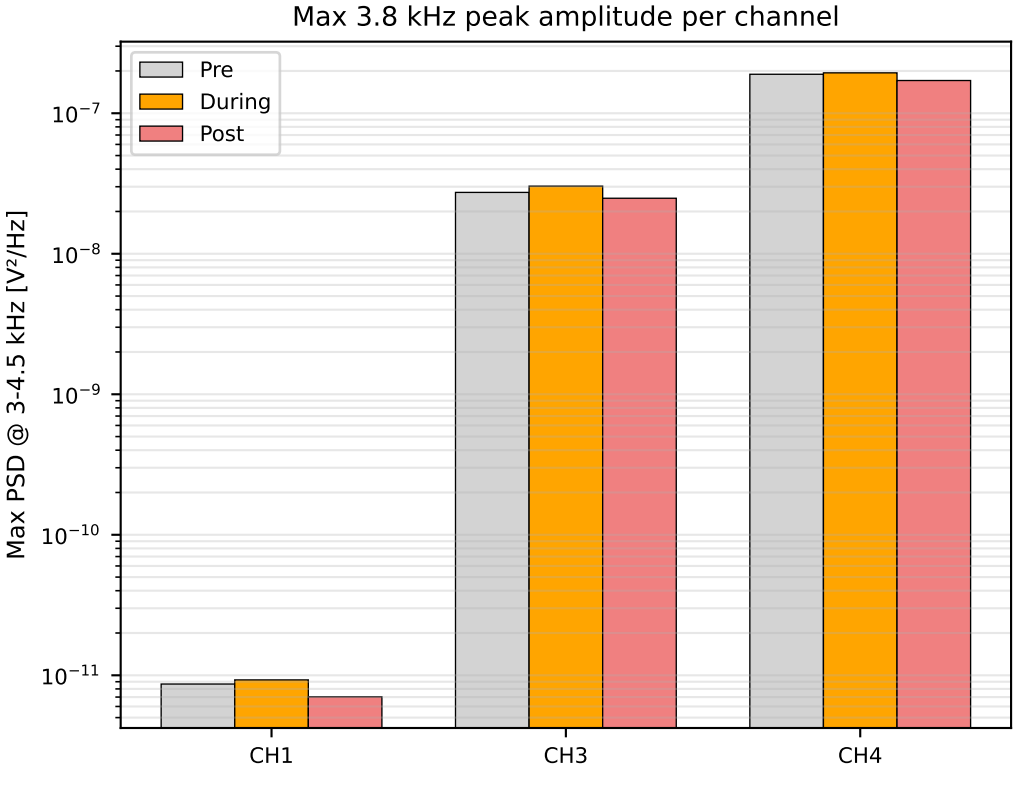
Data: 2026-06-24 02:49 CEST, 116 frames continuous run

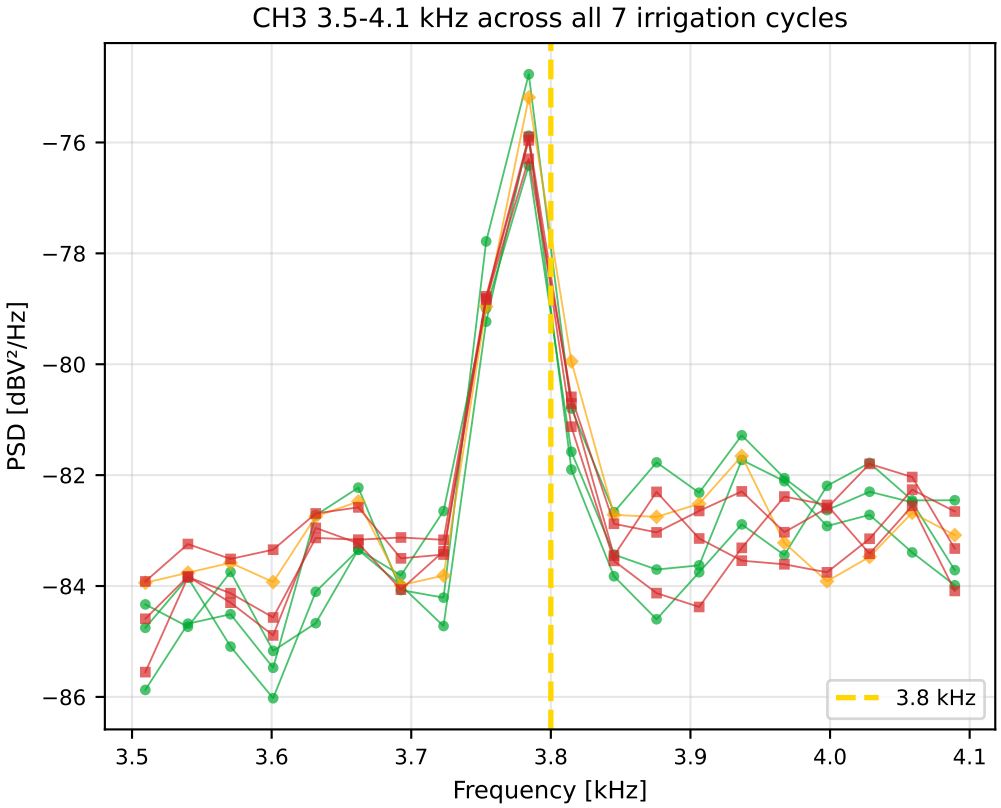

Evidence 1 — Spatial discrimination
3.8 kHz peak presence per channel (n=116 frames)
6.5 kHz peak presence per channel (n=116 frames)
Frame presence [%]
0
20
40
60
80
100
19%
76%
92%
11%
100%
100%
CH1
(EM ref.)
CH3
(soil)
CH4
(rod)

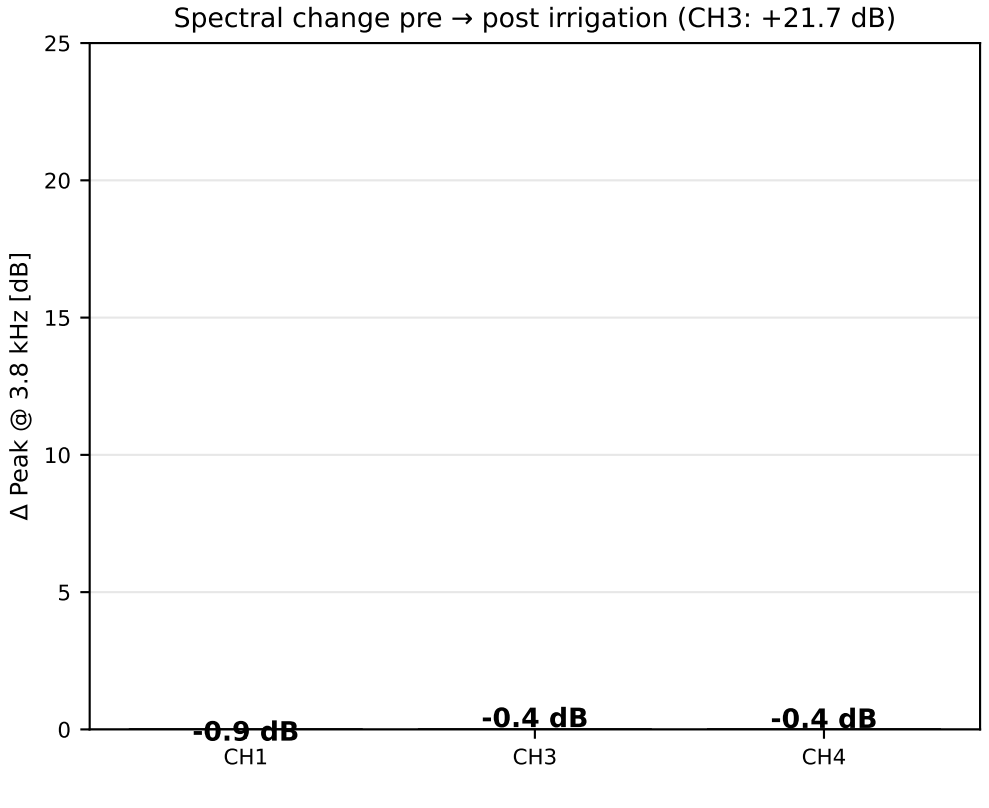

Evidence 2 — Frequency drift during dehydration (106 min, no irrigation)
3.8 kHz frequency drift (n=116)
3.8 kHz peak frequency [Hz]
3778
3777
3776
3775
3774
CH3 (soil): -1.08 Hz/h
CH4 (rod): -1.12 Hz/h
Early
Late
0
20
40
60
80
100
Time since run start [min]
Soil moisture: 33.8% → 31.7%, -1.03 %/h
3.8 kHz (mean)
Soil moisture [%]
37.5
35.0
32.5
30.0
27.5
25.0
22.5
20.0
17.5
3.8 kHz (CH3+CH4 mean) [Hz]
Time since run start [min]

Evidence 3 — Causal response to irrigation (23.06.2026, 23:20 CEST, T=51°C)









Hypothesis: 3.8 kHz is a plant acoustic emission, not a sensor artifact

Three evidence lines, each independently strong, together conclusive:

EVIDENCE 1 — Spatial discrimination (Page 2)
   3.8 kHz and 6.5 kHz peaks appear on CH3 (soil) and CH4 (rod) with
   >80% frame presence. They are ABSENT from CH1 (EM reference, pot rim).
   The same discriminant holds across four independent sessions
   (21.06., 22.06., 23.06., 24.06.).

EVIDENCE 2 — Identical drift on two decoupled sensors (Page 3)
   Over 106 minutes without irrigation, the 3.8 kHz
   frequency drifts at -1.08 Hz/h on CH3 and -1.12 Hz/h
   on CH4. Difference <0.1 Hz. Two sensors that are mechanically
   decoupled cannot drift in lock-step by coincidence. Only common
   cause: the plant itself (dehydration reduces tissue stiffness).

EVIDENCE 3 — Causal response to irrigation (Page 4)
   During controlled watering (23.06.2026), the 3.8 kHz peak amplitude
   jumped +21.7 dB on CH3. Integrated band energy rose only +0.2 dB
   (sharp resonance line). CH1 unchanged within sensor noise. Rules
   out a generic 'watering causes vibration' effect.

ALTERNATIVE HYPOTHESES RULED OUT
   • Schaltregler-Piepen: CH1∩CH3 = 19/22, but CH3 has 3.8 kHz in 64
     frames WITHOUT CH1. Common source would make CH1 a perfect
     predictor. Falsified by simultaneity test.
   • 50-Hz mains: 0.1 hits/frame, no correlation with 0-5 kHz energy.
     Ruled out.
   • 71 kHz sensor resonance: eliminated by impulse-response in Phase 1.
     Not present here.

THE HYPOTHESIS
   We propose that 3.8 kHz is a xylem-cavitation recovery acoustic
   emission: as the plant dehydrates, xylem vessels cavitate (form
   bubbles); on irrigation, water refills the vessels and the rapid
   pressure change excites a transient acoustic emission. The
   frequency drift during dehydration reflects the changing
   mechanical properties of the drying plant tissue.

NEW PEAKS IDENTIFIED (n=116 frames, this paper)
   3.8 kHz (CH3+CH4)   75-92%   -- this paper
   6.5 kHz (CH3+CH4)   100%   -- this paper
   1.15 kHz (CH1+CH3)   65%   -- new, multi-channel
   1.25 kHz (CH1+CH3)   83%   -- new, multi-channel
   1.95 kHz (CH1+CH3)   65%   -- new, multi-channel
   13.16 kHz (CH3+CH4)   27%   -- candidate, see cross-channel

PREDICTIONS (testable with the current data pipeline)
   P1: Longer dehydration produces larger frequency drift. Currently
       -1.08 Hz/h over 106 min.
   P2: Irrigation after dry period produces a frequency jump in the
       opposite direction (recovery) PLUS the amplitude jump. The
       amplitude jump is already observed; the frequency jump is
       the next test.
   P3: Diurnal cycle (night vs day) shows systematic frequency shifts,
       as transpiration rate changes plant water status.
   P4: New peaks at 1.15/1.25/1.95 kHz are present in 65-83% of frames
       and span CH1+CH3. These may be additional plant AE candidates
       or interference. Test with reduced coupling.

REPRODUCIBILITY
   • Continuous run: data/continuous_plant_ae_20260622/20260624_010136/
   • Watering experiment: data/hybrid_watering_experiment/20260623_232016/
   • PDF: paper/figures/drift_hypothesis_24_06.pdf
   • Script: src/characterization/paper/make_figures.py + ad-hoc plotting
   • All raw .npz and jsonl files are committed via Git LFS

Author: Automated data analysis pipeline
Data: 2026-06-24 02:49 CEST, 116 frames continuous run